Methods Paper - Chapter 1

Single-cell RNA-seq preprocessing, clustering, bibliography annotation, and subcluster marker inspection

SingleCell Pipeline

2026-07-15

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This methods-paper R Markdown documents the current Chapter 1 workflow only. It is intentionally safe to knit: code blocks are displayed but not executed. To run the analysis, execute the corresponding sections from `workflow/capitulo1_single_cell.Rmd` inside the Docker container.

The active annotation path is bibliography-based (`biblio_marks_custom.txt` -> `pbmc_harmony$celltype`). Reference-transfer annotation is not part of this draft. The document stops after the optional subcluster marker-inspection section; h5ad export, pseudobulk, DESeq2, GO, network, and pseudotime analyses are intentionally excluded for now.

0.1 Minimum computing requirements

The table below summarizes the computing requirements for running Chapter 1 on a personal computer. These are practical requirements for the downstream Seurat/R workflow after Cell Ranger output has already been produced.

Component	Minimum	Recommended
Operating system	Linux, macOS, or WSL2	Linux server or macOS with Docker Desktop
CPU cores	4	16 or more
RAM	32 GB	64-128 GB for merged multi-sample objects
Disk space	100 GB free	500 GB or more when keeping FASTQ, Cell Ranger output, and R objects
Software	Docker	Docker plus external Cell Ranger installation
Inputs	Cell Ranger `filtered_feature_bc_matrix/` per sample	Raw FASTQ plus Cell Ranger output retained for provenance

Small pilot datasets can be run on a laptop or desktop if memory is sufficient. For full experiments, repeated clustering sweeps, Cell Ranger processing from FASTQ, or large merged Seurat objects, a server or external cluster is strongly recommended. Cell Ranger is CPU- and memory-intensive and is not installed inside the downstream `matigara/scrnaseq:latest` Docker image.

0.2 Materials and data preparation

The workflow is designed to run inside the Docker image `matigara/scrnaseq:latest`, with the project directory mounted as `/workspace`. In the current server setup, the relevant host paths are:

```
/projects/aivvy/SingleCell/workflow/capitulo1_single_cell.Rmd
/projects/aivvy/SingleCell/workflow/ScRNA_Analysis_Functions.R
/projects/aivvy/resultados/
```

0.2.1 Minimal components required to run the pipeline

The minimum inputs for Chapter 1 are:

1. A working Docker installation.
2. The SingleCell workflow files and helper functions.
3. One Cell Ranger `filtered_feature_bc_matrix/` directory per sample.
4. A bibliography marker table with columns `cell types` and `gene`.
5. Enough local or server memory to hold the merged Seurat object.

For this run, the marker table used by the workflow is:

```
/workspace/biblio_marks_custom.txt
```

The marker-table order is biologically meaningful. The dotplot helper preserves marker-table row order on the Y axis, while the X axis is ordered according to the cell-type order implied by that same marker table. Labels with numeric suffixes, such as `Epidermis Hypocotyl.1` and `Epidermis Hypocotyl.2`, are kept adjacent to their parent marker category instead of being moved to the end.

0.2.2 Docker startup

For a new local run, start the container from the host folder that contains the data, workflow, and output directory:

```
docker run -it --rm \
  -v ~/your-workdir:/workspace \
  matigara/scrnaseq:latest /bin/bash
```

On the current server, the Docker container is already running and the project is mounted at `/workspace`. From inside the container, start R and run the workflow by sections:

```
R
```

0.2.3 Raw data download and Cell Ranger processing

This Chapter 1 workflow starts from Cell Ranger output, specifically the `outs/filtered_feature_bc_matrix/` folder for each sample. If only raw FASTQ files are available, first download or stage the FASTQs, then run Cell Ranger for each sample before starting the R workflow.

In the archived Chapter 0 run available on the server, raw FASTQ files were not present at execution time. The script therefore validated existing Cell Ranger outputs and copied the Cell Ranger metrics summaries. The validated run used this layout:

```
/workspace/cellranger/Sample_0N/outs/filtered_feature_bc_matrix/
/workspace/cellranger/Sample_05N/outs/filtered_feature_bc_matrix/
/workspace/cellranger/Sample_5N/outs/filtered_feature_bc_matrix/
```

The validation script used for that run is kept in the repository as:

```
workflow/runs/cap0_r1.sh
```

Its logic checks, for each sample, that the following Cell Ranger files exist and are non-empty:

```
outs/filtered_feature_bc_matrix/matrix.mtx.gz
outs/filtered_feature_bc_matrix/barcodes.tsv.gz
outs/filtered_feature_bc_matrix/features.tsv.gz
outs/metrics_summary.csv
```

For a new analysis starting from public FASTQ files, first create a FASTQ staging directory and download the reads. The exact accession list depends on the experiment, but the workflow expects one folder per sample after download. Common download patterns are:

```
# Option A: ENA/URL download when direct FASTQ links are available
mkdir -p /workspace/fastq/ScWT
wget -c -P /workspace/fastq/ScWT URL_TO_R1_FASTQ.gz
wget -c -P /workspace/fastq/ScWT URL_TO_R2_FASTQ.gz

# Option B: SRA Toolkit when SRR accessions are provided
prefetch SRR_ACCESSION
fasterq-dump SRR_ACCESSION --split-files --threads 8 --outdir /workspace/fastq/ScWT
gzip /workspace/fastq/ScWT/*.fastq
```

After FASTQs are available, run Cell Ranger for each sample. Cell Ranger is not bundled inside the Docker image used for the downstream R/Python workflow, so it should be run on the host, a server, or an HPC environment where Cell Ranger and the appropriate reference transcriptome are installed.

Generic Cell Ranger command structure:

```
cellranger count \
  --id ScWT \
  --transcriptome /path/to/reference_transcriptome \
  --fastqs /path/to/fastq_directory \
  --sample ScWT \
  --localcores 16 \
  --localmem 128
```

Use the resulting `filtered_feature_bc_matrix/` path in the sample manifest in Section 1.

0.2.4 Cluster or server requirement

A cluster is not strictly required for small pilot datasets, but server or HPC execution is recommended once the number of cells, samples, or parameter sweeps increases. The most memory-intensive steps are merging samples, scaling, PCA/UMAP, neighbor construction, marker detection, and subclustering. Cell Ranger also benefits from multiple CPU cores and high memory, especially when processing raw FASTQ files.

0.3 Configuration

Edit this block before running the pipeline.

```
# CONFIGURATION - Edit only this block before running the pipeline
# ============================================================================

# Directory containing the pipeline helper scripts.
# Inside the Docker container this is typically /workspace/SingleCell
PIPELINE_DIR <- "/workspace/SingleCell/workflow"

# Root directory for your project data and results.
# All result files will be written to DATA_DIR/resultados/<step>/
DATA_DIR   <- "/workspace/."
base_dir   <- file.path(DATA_DIR, "resultados")

# ============================================================================
```

0.4 Initialization

Load helper scripts, set reproducibility options, and create output directories.

```
# INITIALIZATION
# ============================================================================

# -- Load helper scripts ---------------------------------------------------
# Each file is fully documented at https://github.com/clifford2001/SingleCell
source(file.path(PIPELINE_DIR, "load_libraries.R"))          # all R packages
source(file.path(PIPELINE_DIR, "ScRNA_Analysis_Functions.R"))# analysis functions

set.seed(1807)
options(Seurat.allow.s4 = FALSE)  # required for Seurat 5 compatibility
setwd(DATA_DIR)

# -- Create per-step output directories ------------------------------------
list2env(create_pipeline_dirs(base_dir), envir = .GlobalEnv)  # creates output folders and loads their

# output_dir is the global variable used by save_pdf / save_qc / save_vln helpers.
# It is reassigned at the start of each section to the appropriate step directory.
output_dir <- base_dir


# ============================================================================
# ########################  PART 1 - SINGLE-CELL ANALYSIS  ###################
# ============================================================================
```

0.5 Section 1 - DATA LOADING AND PRE-FILTER QC

```
# SECTION 1 - DATA LOADING AND PRE-FILTER QC
# ============================================================================
# Each sample is loaded from its input file and mitochondrial / chloroplast
# read fractions are computed per cell to guide filtering thresholds in
# Section 2. Pre-filter violin plots are saved to 01_qc/.
#
# +- CHANGE FOR YOUR ORGANISM ------------------------------------------------
#   Arabidopsis : mt_pattern = "^ATMG"  |  cp_pattern = "^ATCG"
#   Human       : mt_pattern = "^MT-"   |  cp_pattern = NULL
#   Mouse       : mt_pattern = "^mt-"   |  cp_pattern = NULL
# +---------------------------------------------------------------------------
output_dir <- dir_01

# -- Sample manifest (CellRanger filtered_feature_bc_matrix) ------------------
# Add one entry per sample. Each entry needs:
#   file      - path to the filtered_feature_bc_matrix/ directory (relative to DATA_DIR)
#   label     - unique name for this sample (appears in all plots)
#   condition - experimental group this sample belongs to
samples <- list(
  list(file = "cellranger_v2/ScWT/outs/filtered_feature_bc_matrix",  label = "WT",  condition = "WT"),
  list(file = "cellranger_v2/Scpifq/outs/filtered_feature_bc_matrix", label = "pifq", condition = "pifq


# -- Plot colors (one color per sample label) ---------------------------------
colors <- c(
  "WT"  = "#66c2a5",
  "pifq"  = "#fc8d62"
)


mt_pattern <- "^ATMG"  # Arabidopsis mitochondrial genes
cp_pattern <- "^ATCG"  # Arabidopsis chloroplast genes


# Load all samples using the helper function
seurat_list_raw <- load_seurat_samples(samples = samples,
                                       DATA_DIR = DATA_DIR,
                                       mt_pattern = mt_pattern,
                                       cp_pattern = cp_pattern)

plot_qc_batch(seurat_list_raw, colors, "qc_prefilter.pdf")


# ============================================================================

message("\nOK SECTION 1 COMPLETE: QC pre-filter plots saved")
```

Representative output from Section 1:

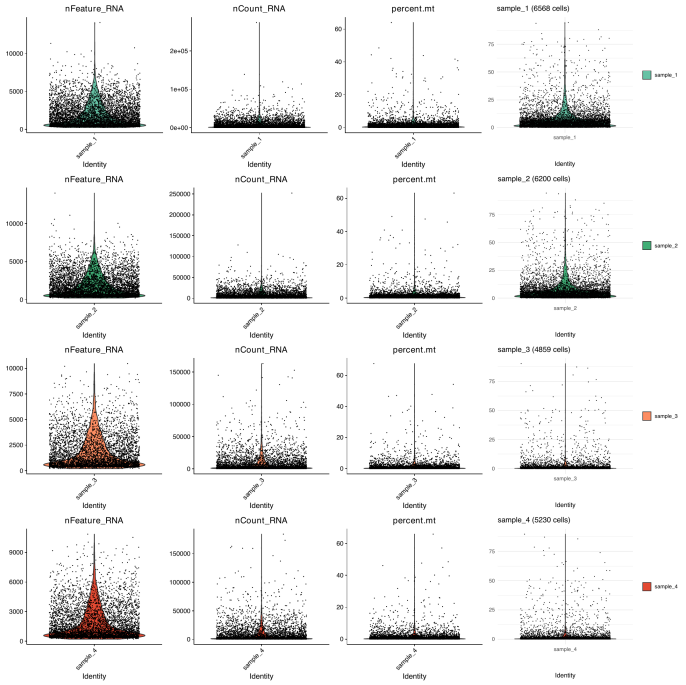


Figure 1: Pre-filter QC violin plots.

0.6 Section 2 - CELL FILTERING AND DOUBLET DETECTION

```
# SECTION 2 - CELL FILTERING AND DOUBLET DETECTION
# ================================================================================
# Thresholds are set based on the pre-filter violin plots (01_qc/qc_prefilter.pdf).
# Cells below min_features or above max_mt are removed. DoubletFinder is
# applied per sample to discard putative doublets.
#
# +- ADJUST THRESHOLDS AFTER INSPECTING 01_qc/qc_prefilter.pdf -----------------
#   min_features : minimum number of detected genes per cell (default 200)
#   max_mt       : maximum mitochondrial read percentage  (default 5 %)
# +----------------------------------------------------------------------------
output_dir <- dir_01   # both pre- and post-filter QC plots go to 01_qc/

seurat_list <- filter_seurat_samples(seurat_list_raw, min_features = 200, max_mt = 5)

plot_qc_batch(seurat_list, colors, "qc_postfilter.pdf")

# Checkpoint - restore with: seurat_list <- readRDS("resultados/objects/seurat_list_postfilter.rds")
saveRDS(seurat_list, file.path(dir_objects, "seurat_list_postfilter.rds"))


# ================================================================================

message("\nOK SECTION 2 COMPLETE: Filtering and doublet detection complete")
```

Representative output from Section 2:

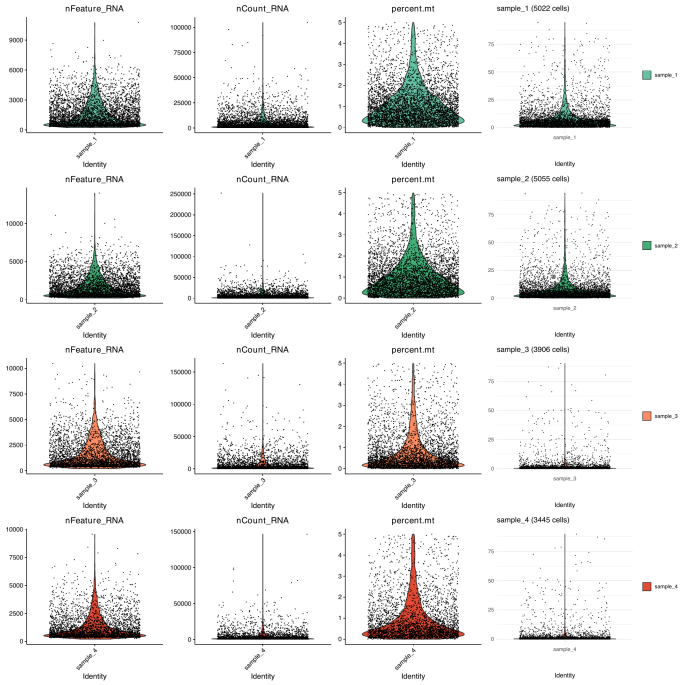


Figure 2: Post-filter QC violin plots.

0.7 Section 3 - MERGE AND INITIAL PREPROCESSING

```
# SECTION 3 - MERGE AND INITIAL PREPROCESSING
# ============================================================================
# Filtered samples are merged and preprocessed: log-normalization, variable
# feature selection (VST, 2,000 features), scaling, PCA (30 PCs), and UMAP.
# The resulting UMAP shows batch effects before integration.

output_dir <- dir_01

pbmc_harmony <- reduce(seurat_list, merge) %>%  # merge all samples into one object
  NormalizeData(verbose = FALSE) %>%
  FindVariableFeatures(selection.method = "vst", nfeatures = 2000, verbose = FALSE) %>%
  ScaleData(verbose = FALSE) %>%
  RunPCA(npcs = 30, verbose = FALSE) %>%
  RunUMAP(reduction = "pca", dims = 1:30, verbose = FALSE)

save_pdf(DimPlot(pbmc_harmony, group.by = "orig.ident", cols = colors),
         "umap_preharmony.pdf")

# Checkpoint - restore with: pbmc_harmony <- readRDS("resultados/objects/pbmc_harmony_preharmony.rds")
saveRDS(pbmc_harmony, file.path(dir_objects, "pbmc_harmony_preharmony.rds"))


# ============================================================================

message("\nOK SECTION 3 COMPLETE: Merge and preprocessing complete")
```

Representative output from Section 3:

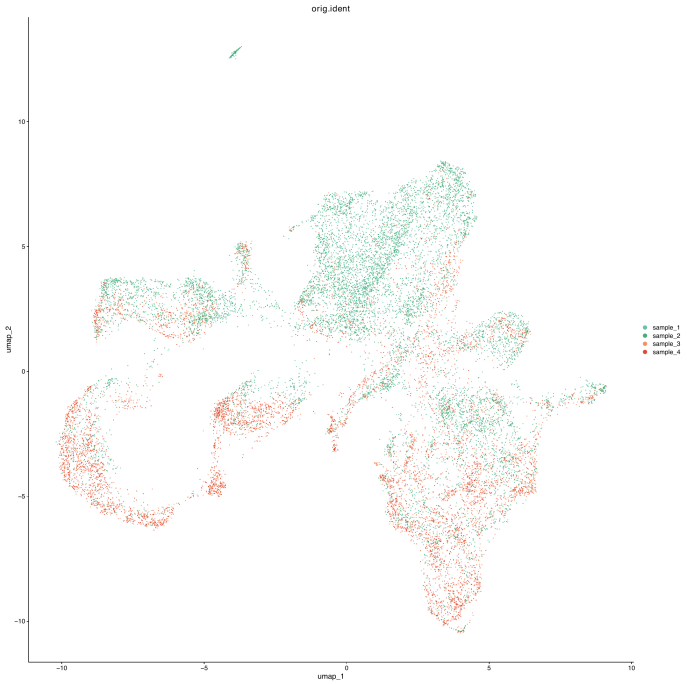


Figure 3: UMAP before Harmony correction, colored by sample.

0.8 Section 4 - HARMONY BATCH CORRECTION

```
# SECTION 4 - HARMONY BATCH CORRECTION
# ============================================================================
# Harmony adjusts cell embeddings to remove sample-level batch effects while
# preserving biological variation. All downstream steps use the "harmony"
# reduction instead of "pca".

output_dir <- dir_01

pbmc_harmony <- pbmc_harmony %>%
  RunHarmony("orig.ident", plot_convergence = FALSE) %>%
  RunUMAP(reduction = "harmony", dims = 1:30, verbose = FALSE)

save_pdf(DimPlot(pbmc_harmony, group.by = "orig.ident", cols = colors),
         "umap_postharmony.pdf")

# Checkpoint - restore with: pbmc_harmony <- readRDS("resultados/objects/pbmc_harmony_postharmony.rds")
saveRDS(pbmc_harmony, file.path(dir_objects, "pbmc_harmony_postharmony.rds"))


# ============================================================================

message("\nOK SECTION 4 COMPLETE: Harmony integration complete")
```

Representative output from Section 4:

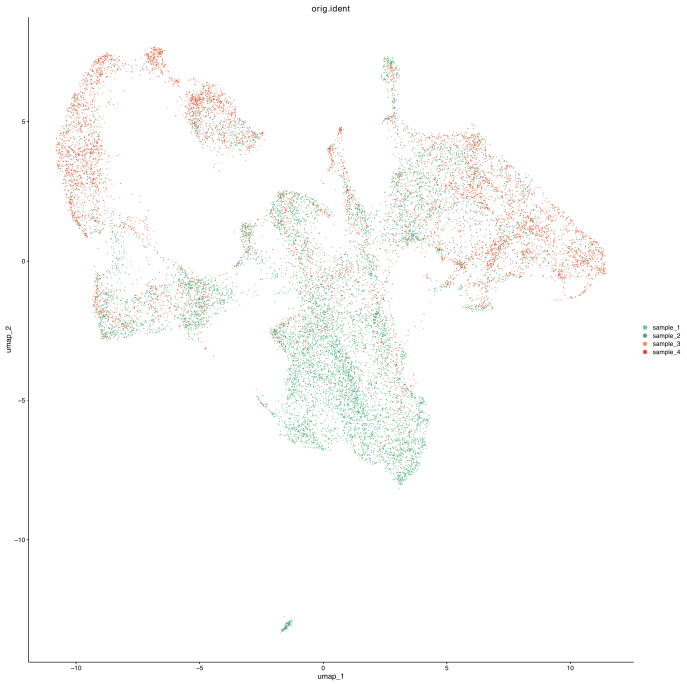


Figure 4: UMAP after Harmony correction, colored by sample.

0.9 Section 5 - RESOLUTION OPTIMIZATION

```
# SECTION 5 - RESOLUTION OPTIMIZATION
# ============================================================================
# Clustree tracks how Leiden communities split or merge across candidate
# resolutions. Choose the lowest resolution where clusters stabilize.
#
# +- PARAMETERS ------------------------------------------------------------
#   resolutions_test : Leiden resolutions swept by clustree
#   -> Inspect elbow_plot.pdf and clustree2.pdf before setting cluster_resolution in Section 6.
# +-------------------------------------------------------------------------
resolutions_test <- c(0.15, 0.35, 0.45, 0.55, 1.0)

output_dir <- dir_02

# Elbow plot: k-means within-cluster sum of squares across candidate k values.
# This is only a diagnostic plot; it does not change the clustering parameters.
k_range <- 1:31
pca_data <- Embeddings(pbmc_harmony, "pca")[, 1:30]
wss <- sapply(
  k_range,
  function(k) kmeans(pca_data, centers = k, nstart = 4)$tot.withinss
)

elbow_plot <- ggplot(data.frame(k = k_range, wss = wss), aes(k, wss)) +
  geom_line() +
  geom_point() +
  labs(
    x = "Number of clusters (k)",
```

```
    y = "Within-cluster sum of squares"
  ) +
  theme_minimal()

save_pdf(elbow_plot, "elbow_plot.pdf", w = 18, h = 18)

clu <- pbmc_harmony %>%
  RunUMAP(reduction = "harmony", dims = 1:30, verbose = FALSE) %>%
  FindNeighbors(reduction = "harmony", dims = 1:30,
                k.param = 20, verbose = FALSE)

for (res in resolutions_test)
  clu <- FindClusters(clu, resolution = res, algorithm = 4, verbose = FALSE)

save_pdf(clustree(clu, prefix = "RNA_snn_res."), "clustree2.pdf", w = 18, h = 18)


# ============================================================================

message("\nOK SECTION 5 COMPLETE: Elbow plot and clustree saved - inspect elbow_plot.pdf and clustree2.p
```

Representative outputs from Section 5:

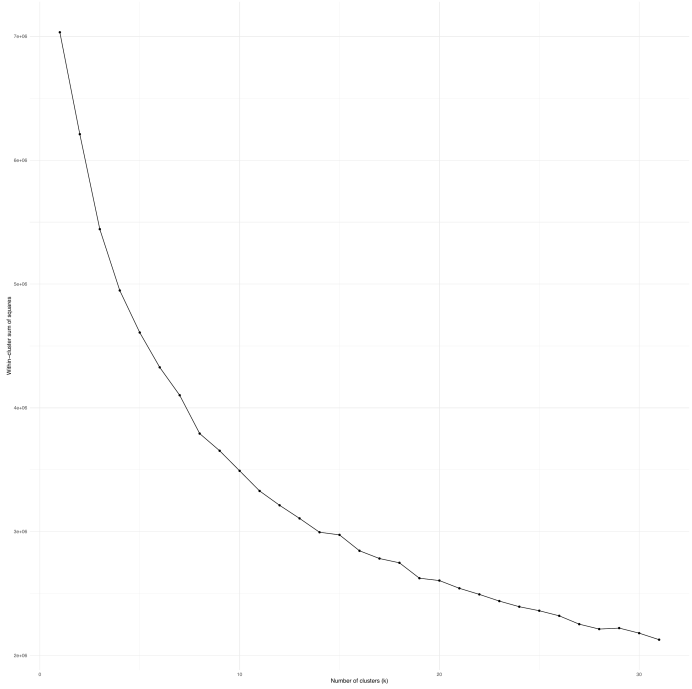


Figure 5: Elbow diagnostic used to inspect candidate cluster numbers.

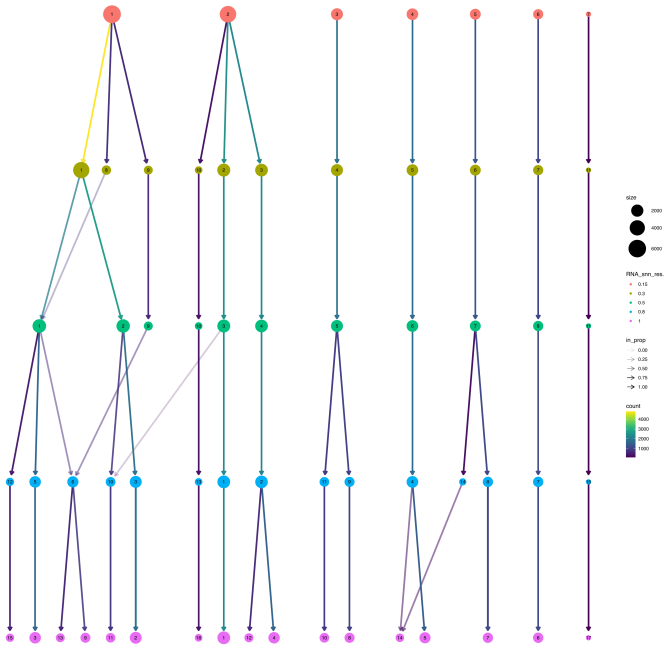


Figure 6: Cluster tree across candidate Leiden resolutions.

0.10 Section 6 - FINAL CLUSTERING

```
# SECTION 6 - FINAL CLUSTERING
# ============================================================================
# Apply the selected resolution for the final cluster assignment.
# After clustering, a UMAP coloured by Seurat cluster and a bar chart of
# cells per sample are saved to 02_clustering/.
#
# +- SET RESOLUTION AFTER INSPECTING clustree.pdf ----------------------------
#   cluster_resolution : Leiden resolution for final clustering (default 0.3)
# +--------------------------------------------------------------------------
cluster_resolution <- 0.35
output_dir <- dir_02

# clu (Section 5) was temporary - re-run on pbmc_harmony to embed the final clusters
pbmc_harmony <- pbmc_harmony %>%
  RunUMAP(reduction = "harmony", dims = 1:30, verbose = FALSE) %>%
  FindNeighbors(reduction = "harmony", dims = 1:30,
                k.param = 20, verbose = FALSE) %>%
  FindClusters(resolution = cluster_resolution, algorithm = 4, verbose = FALSE)

Idents(pbmc_harmony) <- "seurat_clusters"
save_pdf(DimPlot(pbmc_harmony, group.by = "seurat_clusters", label = TRUE),
         "umap_seuratclusters.pdf")


# ============================================================================

message("\nOK SECTION 6 COMPLETE: Final clustering complete")
```

Representative output from Section 6:

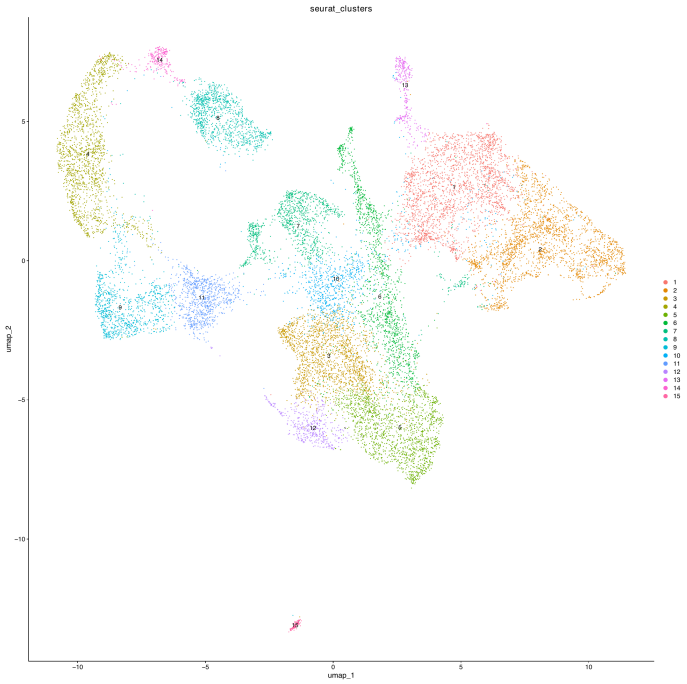


Figure 7: Final UMAP colored by numbered Seurat clusters.

0.11 Section 7 - CELL-TYPE ANNOTATION

```
# SECTION 7 - CELL-TYPE ANNOTATION
# ================================================================================
# Cell-type labels are assigned via bibliography-based markers
# (biblio_marks.txt): the marker table read above is crossed with
# cluster-level differential genes to assign initial labels automatically.
# Result stored in pbmc_harmony$celltype.
#
# A second strategy - reference transfer from a published Seurat object
# (e.g. an Arabidopsis leaf atlas) via FindTransferAnchors/TransferData - is
# not run by default. The full block, plus the downstream sections wired to
# it, is preserved in capitulo1_single_cell_reference_backup.R.

output_dir <- dir_03

biblio_marks_file <- file.path(DATA_DIR, "biblio_marks_custom.txt")
marker_table      <- read.table(biblio_marks_file, header = TRUE, sep = "\t", quote = "")

# -- Bibliography-based annotation ---------------------------------------------
markers <- find_markers(pbmc_harmony,
                        output_file = file.path(output_dir, "FindAllMarkers.tsv"))

pbmc_harmony <- annotate_by_markers(pbmc_harmony, markers,
                                    reference_file = biblio_marks_file)
# Annotation stored in: pbmc_harmony$celltype

plot_marker_dotplot(
  pbmc_harmony,
  marker_table,
  annot_col = "celltype", # uses the newly assigned annotation column
  outfile   = file.path(output_dir, "dotplot_marker_table_annotation_biblio.pdf"),
  width = 18, height = 18
)

save_pdf(DimPlot(pbmc_harmony, group.by = "celltype",
                 label = TRUE, repel = TRUE, raster = FALSE),
         "umap_annotation_biblio.pdf")


# ================================================================================

message("\nOK SECTION 7 COMPLETE: Cell-type annotation complete")
```

Representative outputs from Section 7:

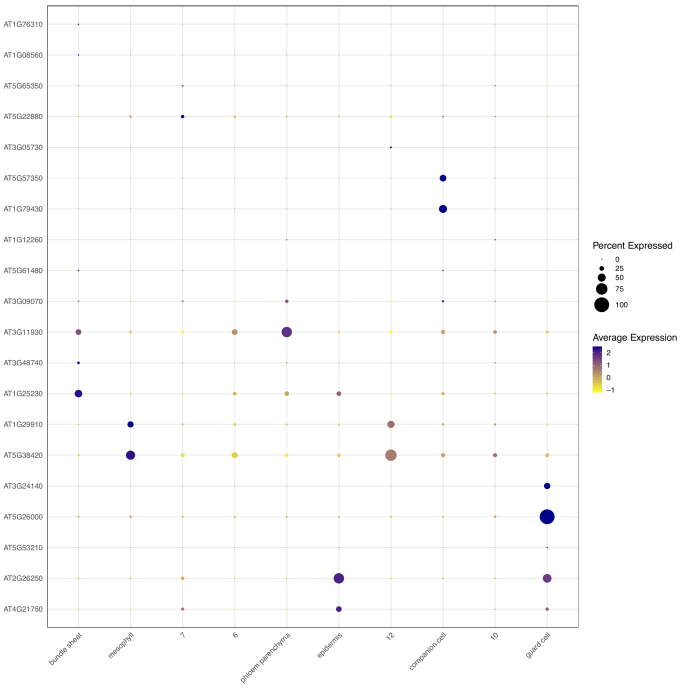


Figure 8: Marker dotplot after bibliography-based cell-type annotation.

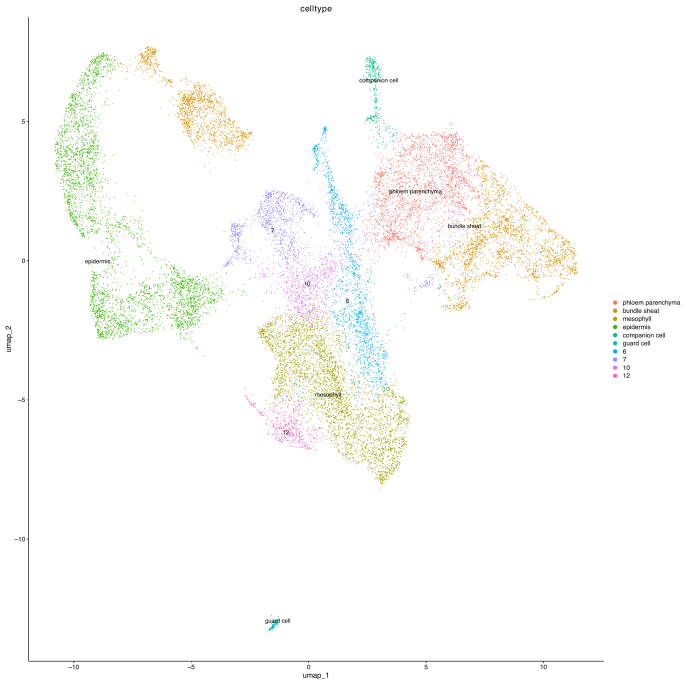


Figure 9: UMAP colored by bibliography-based cell-type annotation.

0.12 Section 8 - ANNOTATED CLUSTREE

```
# SECTION 8 - ANNOTATED CLUSTREE
# ==============================================================================
# Reuses the resolution sweep object from Section 5 and overlays the final
# bibliography-based annotation on each node.

output_dir <- dir_03

# Mode: returns the most frequent annotation in each clustree node.
Mode <- function(x) {
  x <- as.character(x)
  x <- x[!is.na(x) & nzchar(x)]
  if (length(x) == 0) return(NA_character_)
  names(sort(table(x), decreasing = TRUE))[1]
}

stopifnot(exists("clu"))
stopifnot("celltype" %in% colnames(pbmc_harmony@meta.data))

celltype_label <- as.character(pbmc_harmony$celltype)
names(celltype_label) <- Cells(pbmc_harmony)
clu$celltype_label <- celltype_label[Cells(clu)]

print(table(clu$celltype_label, useNA = "ifany"))

save_pdf(
  clustree(clu, prefix = "RNA_snn_res.",
           node_label = "celltype_label", node_label_aggr = "Mode"),
  "clustree_annotated.pdf", w = 14, h = 14
)


# ==============================================================================

message("\nOK SECTION 8 COMPLETE: Annotated clustree saved")
```

Representative output from Section 8:

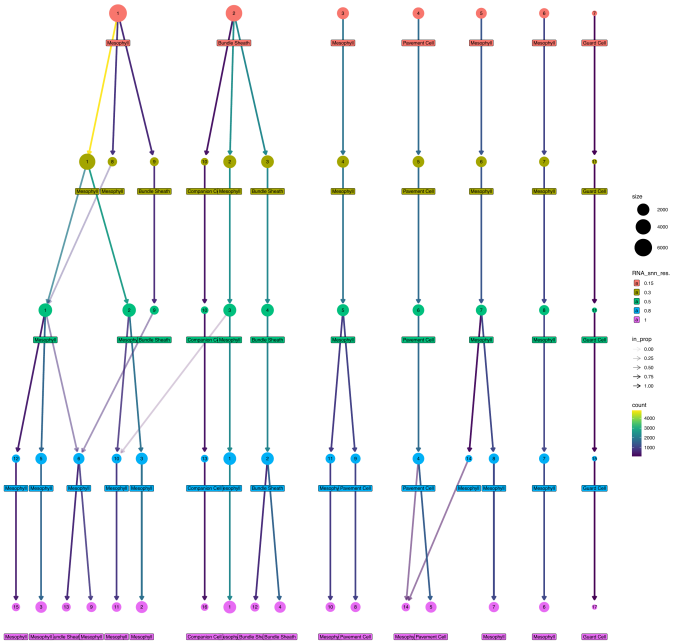


Figure 10: Cluster tree labeled by the dominant bibliography-based annotation in each node.

0.13 Section 9 - GENE EXPRESSION VISUALIZATION

```
# SECTION 9 - GENE EXPRESSION VISUALIZATION
# ============================================================================
# Simple inspection for one gene: violin plot across all annotated cell types
# and UMAP expression overlay.
#
# +- SET GENE TO INSPECT ---------------------------------------------------
#   gene : single gene to inspect
# +-------------------------------------------------------------------------
gene <- "AT5G26000"

output_dir <- dir_04

# JoinLayers is required in Seurat 5 after merging samples.
pbmc_harmony <- JoinLayers(pbmc_harmony)
Idents(pbmc_harmony) <- "celltype"

save_vln(
  VlnPlot(pbmc_harmony, features = gene),
  "vln_gene_all.pdf"
)

save_pdf(
  FeaturePlot(pbmc_harmony, features = gene),
  "feature_gene_all.pdf"
)

# ============================================================================
```

```r
message("\nOK SECTION 9 COMPLETE: Gene expression visualization saved")
```

Representative outputs from Section 9:

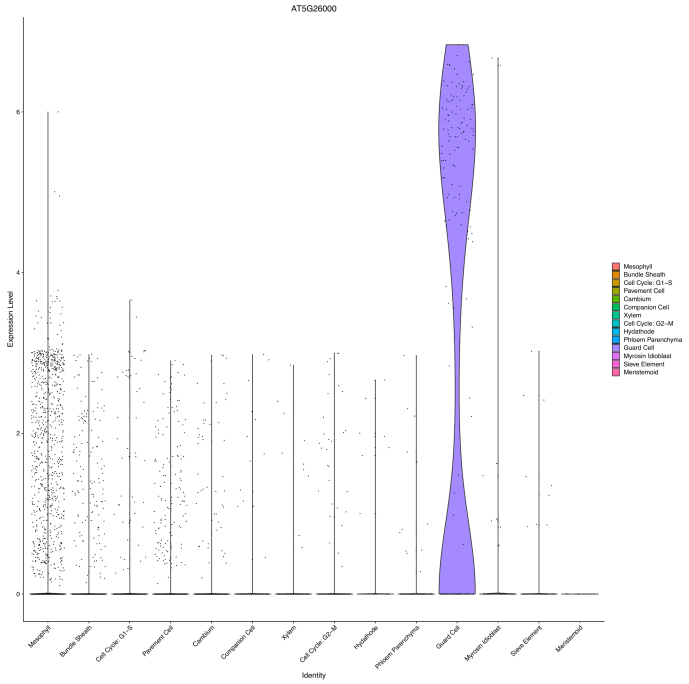


Figure 11: Violin plot for the selected gene across all annotated cell types.

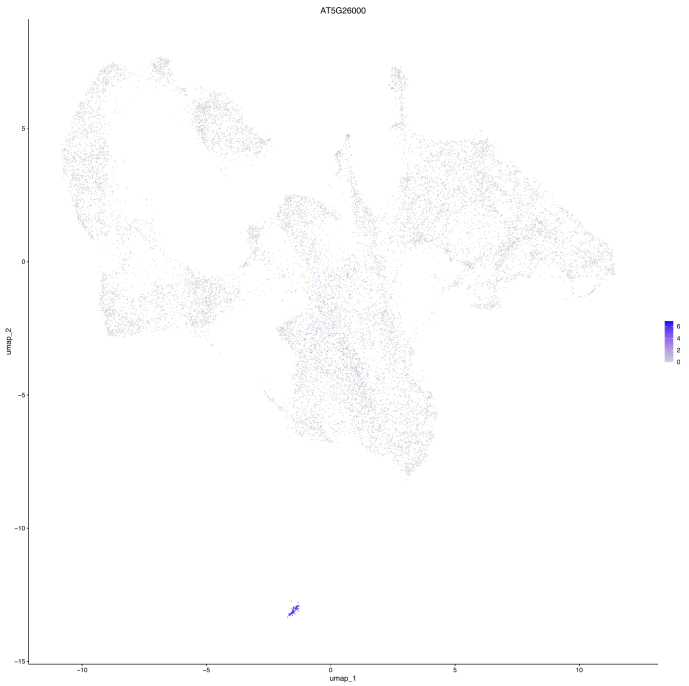


Figure 12: UMAP expression overlay for the selected gene.

0.14 Section 10 - CELL-TYPE GROUPING [OPTIONAL]

```
# SECTION 10 - CELL-TYPE GROUPING  [OPTIONAL]
# ============================================================================
# Fine-grained labels are collapsed into broader categories for downstream
# analyses. Cell types NOT listed in 'grouping' keep their original label.
# Skip this section if you do not need to merge cell types.
#
# +- EDIT THIS MAP TO MATCH YOUR CELL TYPES ---------------------------------
#   Left side  : original label (must match exactly)
#   Right side : new broader label to assign
# +-------------------------------------------------------------------------
# Leave empty if you do not want to merge labels.
#
# Example: uncomment and edit this if Section 7 produced several labels that
# should be treated as one broader group downstream.
#
# grouping <- c(
#   "Epidermis Hypocotyl.1" = "Epidermis Hypocotyl",
#   "Epidermis Hypocotyl.2" = "Epidermis Hypocotyl",
#   "Epidermis Hypocotyl.3" = "Epidermis Hypocotyl",
#   "ENDO1" = "Endodermis",
#   "ENDO2" = "Endodermis",
#   "PHLOEM AND SIEVE ELEMENT" = "PHLOEM"
# )
#
# Current default: no grouping; celltype_grouped is a straight pass-through
# of celltype.
grouping <- c()
```

```r
output_dir <- dir_05

# !!! unpacks the grouping vector as named arguments to recode()
# recode() errors on an empty replacement list, so an empty `grouping` (no
# merging needed) falls back to a straight pass-through instead.
if (length(grouping) > 0) {
  pbmc_harmony$celltype_grouped <- recode(pbmc_harmony$celltype, !!!grouping)
} else {
  pbmc_harmony$celltype_grouped <- pbmc_harmony$celltype
}

save_pdf(
  DimPlot(pbmc_harmony, group.by = "celltype_grouped",  # grouped broad cell types
          label = TRUE, repel = TRUE, raster = FALSE),
  "umap_grouped.pdf"
)


# ============================================================================

message("\nOK SECTION 10 COMPLETE: Cell-type grouping complete")
```

Representative output from Section 10:

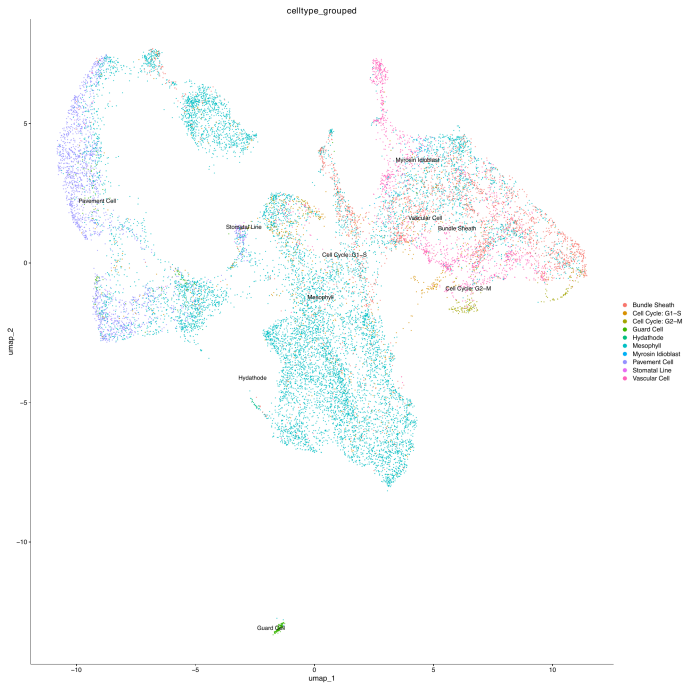


Figure 13: UMAP colored by grouped cell-type labels.

0.15 Section 11 - SUBCLUSTER MARKER INSPECTION [OPTIONAL]

This section is intentionally interactive. It is used to test marker expression against subclusters before deciding whether manual curation is needed. The section does not modify `pbmc_harmony` by default. Individual subcluster UMAPs are saved in a wide format so that UMAP dimensions keep the same visual proportions and do not appear compressed in the report.

```
# SECTION 11 - SUBCLUSTER MARKER INSPECTION  [OPTIONAL]
# ============================================================================
# !!! WARNING: Run this section interactively, step by step.
# !!! Do NOT source the entire script with this section active.
# Skip this section if you are satisfied with the annotation from Section 7.
#
# Purpose: pick one or more annotated cell types, subcluster them, and inspect
# marker expression across the subclusters before deciding whether any manual
# reassignment is needed. This section does NOT modify pbmc_harmony by default.

output_dir   <- dir_05
curation_col <- "celltype"
Idents(pbmc_harmony) <- curation_col

print(table(pbmc_harmony[[curation_col]], useNA = "ifany"))

# Edit this vector using exact names from the table above.
# Example options:
#   types_to_inspect <- c("Epidermis Hypocotyl.1")
#   types_to_inspect <- c("Epidermis Hypocotyl.1", "Epidermis Hypocotyl.2")
#   types_to_inspect <- names(table(pbmc_harmony[[curation_col]]))  # all labels; can be slow

types_to_inspect <- c("2","4","5","8")

subcluster_resolution <- 0.3
subcluster_dims       <- 1:20

subcluster_list    <- list()
inspection_entries <- list()

for (ct in types_to_inspect) {
  safe_name <- make.names(ct)
  file_tag  <- gsub("[^A-Za-z0-9]+", "_", ct)

  message("\nInspecting: ", ct)

  sub_obj <- subcluster_cell_type(
    pbmc_harmony,
    tipo       = ct,
    annot_col  = curation_col,
    resolution = subcluster_resolution,
    dims       = subcluster_dims
  )

  subcluster_list[[safe_name]] <- sub_obj
  print(table(sub_obj$cluster_subtipo, useNA = "ifany"))

  p_dim <- plot_subcluster_umap(sub_obj, ct, output_dir)
```

```
  plot_marker_dotplot(
    sub_obj,
    marker_table,
    annot_col = "cluster_subtipo",
    outfile   = file.path(output_dir, paste0("dotplot_markers_subclusters_", file_tag, ".pdf")),
    width = 18,
    height = 18
  )

  inspection_entries[[safe_name]] <- list(
    umap_plot = p_dim,
    obj       = sub_obj
  )
}

save_subcluster_composite(
  subcluster_list = inspection_entries,
  marker_table    = marker_table,
  output_dir      = output_dir,
  filename        = "subclustering_marker_inspection2.pdf"
)

# After reviewing the PDFs, manually decide whether to reassign subclusters.
# Nothing is changed in pbmc_harmony unless you run an apply_subcluster_reassignment()
# block intentionally.

# Example reassignment template, left commented on purpose:
#
# reassign <- list(
#   Epidermis.Hypocotyl.1 = c(
#     "0"      = "Epidermis Hypocotyl",
#     "1"      = "Epidermis Cotyledon",
#     "others" = "Epidermis Hypocotyl"
#   )
# )
#
# pbmc_harmony <- apply_subcluster_reassignment(
#   obj             = pbmc_harmony,
#   subcluster_list = subcluster_list,
#   reassign        = reassign,
#   source_col      = curation_col,
#   dest_col        = "celltype_curated"
# )
#
# save_pdf(
#   DimPlot(pbmc_harmony, group.by = "celltype_curated",
#           label = TRUE, repel = TRUE, raster = FALSE),
#   "umap_curated.pdf"
# )
# saveRDS(pbmc_harmony, file.path(dir_objects, "pbmc_harmony_curated.rds"))

# ================================================================================
```

```r
message("\nOK SECTION 11 COMPLETE: Subcluster marker inspection saved")
```

Representative output from Section 11:

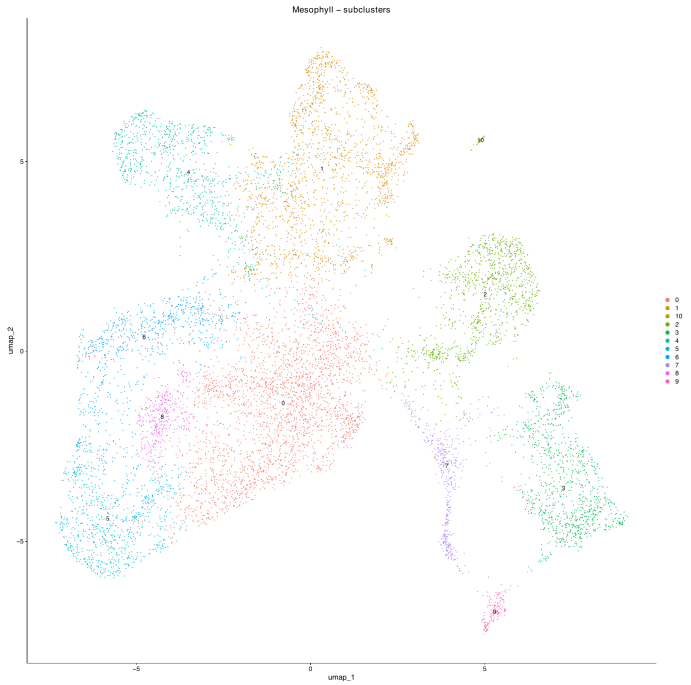


Figure 14: Example subcluster UMAP used for manual marker inspection.

0.16 Current stopping point

This methods draft stops at optional subcluster marker inspection. Export to h5ad, pseudobulk, DESeq2, enrichment, network analysis, and pseudotime are not included in this version.

0.17 Supplementary information

0.17.1 Docker image program and package versions

The following version tables were generated from the active Docker container used to render this methods draft. The complete machine-readable files are kept next to this R Markdown file:

```
reports/docker_cli_versions.tsv
reports/docker_r_package_versions.tsv
reports/docker_python_package_versions.tsv
```

Table 2: Command-line tools detected in the Docker image.

program	version
R	R version 4.5.3 (2026-03-11) – Reassured Reassurer
Rscript	Rscript (R) version 4.5.3 (2026-03-11)
python3	Python 3.12.3
pip	pip 26.1.2 from /opt/venv/lib/python3.12/site-packages/pip (python 3.12)
pandoc	pandoc 3.1.3
cellranger	not installed in Docker image
samtools	not installed in Docker image
bedtools	not installed in Docker image
bowtie2	not installed in Docker image
bwa	not installed in Docker image
STAR	not installed in Docker image
salmon	not installed in Docker image
kallisto	not installed in Docker image
fastqc	not installed in Docker image
multiqc	not installed in Docker image

Table 3: R package inventory in the Docker image, excluding base/recommended R packages.

	Package	Version	Priority	LibPath
1	abind	1.4-8	NA	/usr/local/lib/R/site-library
2	anndataR	1.0.2	NA	/usr/local/lib/R/site-library
3	AnnotationDbi	1.72.0	NA	/usr/local/lib/R/site-library
4	ape	5.8-1	NA	/usr/local/lib/R/site-library
5	aplot	0.2.9	NA	/usr/local/lib/R/site-library
6	askpass	1.2.1	NA	/usr/local/lib/R/site-library
7	assertthat	0.2.1	NA	/usr/local/lib/R/site-library
8	assorthead	1.4.0	NA	/usr/local/lib/R/site-library
9	backports	1.5.1	NA	/usr/local/lib/R/site-library
11	base64enc	0.1-6	NA	/usr/local/lib/R/site-library
12	basilisk	1.22.0	NA	/usr/local/lib/R/site-library
13	batchelor	1.26.0	NA	/usr/local/lib/R/site-library
14	beachmat	2.26.0	NA	/usr/local/lib/R/site-library
15	beeswarm	0.4.0	NA	/usr/local/lib/R/site-library
16	BH	1.90.0-1	NA	/usr/local/lib/R/site-library
17	biglm	0.9-3	NA	/usr/local/lib/R/site-library
18	Biobase	2.70.0	NA	/usr/local/lib/R/site-library
19	BiocGenerics	0.56.0	NA	/usr/local/lib/R/site-library
20	BiocManager	1.30.27	NA	/usr/local/lib/R/site-library

	Package	Version	Priority	LibPath
21	BiocNeighbors	2.4.0	NA	/usr/local/lib/R/site-library
22	BiocParallel	1.44.0	NA	/usr/local/lib/R/site-library
23	BiocSingular	1.26.1	NA	/usr/local/lib/R/site-library
24	BiocVersion	3.22.0	NA	/usr/local/lib/R/site-library
25	Biostrings	2.78.0	NA	/usr/local/lib/R/site-library
26	bit	4.6.0	NA	/usr/local/lib/R/site-library
27	bit64	4.8.2	NA	/usr/local/lib/R/site-library
28	bitops	1.0-9	NA	/usr/local/lib/R/site-library
29	blob	1.3.0	NA	/usr/local/lib/R/site-library
32	BPCells	0.3.1	NA	/usr/local/lib/R/site-library
33	brew	1.0-10	NA	/usr/local/lib/R/site-library
34	brio	1.1.5	NA	/usr/local/lib/R/site-library
35	broom	1.0.13	NA	/usr/local/lib/R/site-library
36	bslib	0.11.0	NA	/usr/local/lib/R/site-library
37	cachem	1.1.0	NA	/usr/local/lib/R/site-library
38	Cairo	1.7-0	NA	/usr/local/lib/R/site-library
39	callr	3.7.6	NA	/usr/local/lib/R/site-library
40	car	3.1-5	NA	/usr/local/lib/R/site-library
41	carData	3.0-6	NA	/usr/local/lib/R/site-library
42	caTools	1.18.3	NA	/usr/local/lib/R/site-library
43	cellranger	1.1.0	NA	/usr/local/lib/R/site-library
44	checkmate	2.3.4	NA	/usr/local/lib/R/site-library
45	circlize	0.4.18	NA	/usr/local/lib/R/site-library
47	classInt	0.4-11	NA	/usr/local/lib/R/site-library
48	cli	3.6.6	NA	/usr/local/lib/R/site-library
49	clipr	0.8.1	NA	/usr/local/lib/R/site-library
50	clue	0.3-68	NA	/usr/local/lib/R/site-library
52	clusterProfiler	4.18.4	NA	/usr/local/lib/R/site-library
53	clustree	0.5.1	NA	/usr/local/lib/R/site-library
55	colorspace	2.1-2	NA	/usr/local/lib/R/site-library
56	commonmark	2.0.0	NA	/usr/local/lib/R/site-library
58	ComplexHeatmap	2.26.1	NA	/usr/local/lib/R/site-library
59	conflicted	1.2.0	NA	/usr/local/lib/R/site-library
60	corrplot	0.95	NA	/usr/local/lib/R/site-library
61	cowplot	1.2.0	NA	/usr/local/lib/R/site-library
62	cpp11	0.5.5	NA	/usr/local/lib/R/site-library
63	crayon	1.5.3	NA	/usr/local/lib/R/site-library
64	credentials	2.0.3	NA	/usr/local/lib/R/site-library
65	crosstalk	1.2.2	NA	/usr/local/lib/R/site-library
66	curl	7.1.0	NA	/usr/local/lib/R/site-library
67	data.table	1.18.4	NA	/usr/local/lib/R/site-library
69	DBI	1.3.0	NA	/usr/local/lib/R/site-library
70	dbplyr	2.5.2	NA	/usr/local/lib/R/site-library
71	DelayedArray	0.36.1	NA	/usr/local/lib/R/site-library
72	DelayedMatrixStats	1.32.0	NA	/usr/local/lib/R/site-library
73	deldir	2.0-4	NA	/usr/local/lib/R/site-library
74	Deriv	4.2.0	NA	/usr/local/lib/R/site-library
75	desc	1.4.3	NA	/usr/local/lib/R/site-library
76	DESeq2	1.50.2	NA	/usr/local/lib/R/site-library
77	devtools	2.5.2	NA	/usr/local/lib/R/site-library
78	diffobj	0.3.6	NA	/usr/local/lib/R/site-library
79	digest	0.6.39	NA	/usr/local/lib/R/site-library

	Package	Version	Priority	LibPath
80	dir.expiry	1.18.0	NA	/usr/local/lib/R/site-library
81	distributional	0.7.0	NA	/usr/local/lib/R/site-library
82	doBy	4.7.1	NA	/usr/local/lib/R/site-library
83	docopt	0.7.2	NA	/usr/local/lib/R/site-library
84	doParallel	1.0.17	NA	/usr/local/lib/R/site-library
85	DOSE	4.4.0	NA	/usr/local/lib/R/site-library
86	dotCall64	1.2	NA	/usr/local/lib/R/site-library
87	DoubletFinder	2.0.6	NA	/usr/local/lib/R/site-library
88	downlit	0.4.5	NA	/usr/local/lib/R/site-library
89	dplyr	1.2.1	NA	/usr/local/lib/R/site-library
90	dqrng	0.4.1	NA	/usr/local/lib/R/site-library
91	dtplyr	1.3.3	NA	/usr/local/lib/R/site-library
92	dynamicTreeCut	1.63-1	NA	/usr/local/lib/R/site-library
93	e1071	1.7-17	NA	/usr/local/lib/R/site-library
94	ellipsis	0.3.3	NA	/usr/local/lib/R/site-library
95	enrichplot	1.30.5	NA	/usr/local/lib/R/site-library
96	enrichR	3.4	NA	/usr/local/lib/R/site-library
97	eulerr	8.0.0	NA	/usr/local/lib/R/site-library
98	evaluate	1.0.5	NA	/usr/local/lib/R/site-library
99	fansi	1.0.7	NA	/usr/local/lib/R/site-library
100	farver	2.1.2	NA	/usr/local/lib/R/site-library
101	fastcluster	1.3.0	NA	/usr/local/lib/R/site-library
102	fastDummies	1.7.6	NA	/usr/local/lib/R/site-library
103	fastmap	1.2.0	NA	/usr/local/lib/R/site-library
104	fastmatch	1.1-8	NA	/usr/local/lib/R/site-library
105	fgsea	1.36.2	NA	/usr/local/lib/R/site-library
106	fields	17.3	NA	/usr/local/lib/R/site-library
107	filelock	1.0.3	NA	/usr/local/lib/R/site-library
108	fitdistrplus	1.2-6	NA	/usr/local/lib/R/site-library
109	FNN	1.1.4.1	NA	/usr/local/lib/R/site-library
110	fontawesome	0.5.3	NA	/usr/local/lib/R/site-library
111	fontBitstreamVera	0.1.1	NA	/usr/local/lib/R/site-library
112	fontLiberation	0.1.0	NA	/usr/local/lib/R/site-library
113	fontquiver	0.2.1	NA	/usr/local/lib/R/site-library
114	forcats	1.0.1	NA	/usr/local/lib/R/site-library
115	foreach	1.5.2	NA	/usr/local/lib/R/site-library
116	forecast	9.0.2	NA	/usr/local/lib/R/site-library
118	formatR	1.14	NA	/usr/local/lib/R/site-library
119	Formula	1.2-5	NA	/usr/local/lib/R/site-library
120	fracdiff	1.5-4	NA	/usr/local/lib/R/site-library
121	fs	2.1.0	NA	/usr/local/lib/R/site-library
122	furrr	0.4.0	NA	/usr/local/lib/R/site-library
123	futile.logger	1.4.9	NA	/usr/local/lib/R/site-library
124	futile.options	1.0.1	NA	/usr/local/lib/R/site-library
125	future	1.70.0	NA	/usr/local/lib/R/site-library
126	future.apply	1.20.2	NA	/usr/local/lib/R/site-library
127	gargle	1.6.1	NA	/usr/local/lib/R/site-library
128	gdtools	0.5.1	NA	/usr/local/lib/R/site-library
129	GeneOverlap	1.46.0	NA	/usr/local/lib/R/site-library
130	generics	0.1.4	NA	/usr/local/lib/R/site-library
131	GENIE3	1.32.0	NA	/usr/local/lib/R/site-library
132	GenomeInfoDb	1.46.2	NA	/usr/local/lib/R/site-library

	Package	Version	Priority	LibPath
133	GenomicRanges	1.62.1	NA	/usr/local/lib/R/site-library
134	gert	2.3.1	NA	/usr/local/lib/R/site-library
135	GetoptLong	1.1.1	NA	/usr/local/lib/R/site-library
136	ggbeeswarm	0.7.3	NA	/usr/local/lib/R/site-library
137	ggdist	3.3.3	NA	/usr/local/lib/R/site-library
138	ggforce	0.5.0	NA	/usr/local/lib/R/site-library
139	ggfun	0.2.0	NA	/usr/local/lib/R/site-library
140	ggiraph	0.9.6	NA	/usr/local/lib/R/site-library
141	ggnewscale	0.5.2	NA	/usr/local/lib/R/site-library
142	ggplot2	4.0.3	NA	/usr/local/lib/R/site-library
143	ggplotify	0.1.3	NA	/usr/local/lib/R/site-library
144	ggpubr	0.6.3	NA	/usr/local/lib/R/site-library
145	ggraph	2.2.2	NA	/usr/local/lib/R/site-library
146	ggrastr	1.0.2	NA	/usr/local/lib/R/site-library
147	ggrepel	0.9.8	NA	/usr/local/lib/R/site-library
148	ggridges	0.5.7	NA	/usr/local/lib/R/site-library
149	ggsci	5.0.0	NA	/usr/local/lib/R/site-library
150	ggsignif	0.6.4	NA	/usr/local/lib/R/site-library
151	ggtangle	0.1.2	NA	/usr/local/lib/R/site-library
152	ggtree	4.0.5	NA	/usr/local/lib/R/site-library
153	ggvenn	0.1.19	NA	/usr/local/lib/R/site-library
154	gh	1.6.0	NA	/usr/local/lib/R/site-library
155	gitcreds	0.1.2	NA	/usr/local/lib/R/site-library
156	GlobalOptions	0.1.4	NA	/usr/local/lib/R/site-library
157	globals	0.19.1	NA	/usr/local/lib/R/site-library
158	glue	1.8.1	NA	/usr/local/lib/R/site-library
159	GO.db	3.22.0	NA	/usr/local/lib/R/site-library
160	goftest	1.2-3	NA	/usr/local/lib/R/site-library
161	googledrive	2.1.2	NA	/usr/local/lib/R/site-library
162	googlesheets4	1.1.2	NA	/usr/local/lib/R/site-library
163	GOSemSim	2.36.0	NA	/usr/local/lib/R/site-library
164	gplots	3.3.0	NA	/usr/local/lib/R/site-library
166	graphlayouts	1.2.3	NA	/usr/local/lib/R/site-library
169	gridExtra	2.3	NA	/usr/local/lib/R/site-library
170	gridGraphics	0.5-1	NA	/usr/local/lib/R/site-library
171	grr	0.9.5	NA	/usr/local/lib/R/site-library
172	gson	0.1.0	NA	/usr/local/lib/R/site-library
173	gtable	0.3.6	NA	/usr/local/lib/R/site-library
174	gtools	3.9.5	NA	/usr/local/lib/R/site-library
175	h5mread	1.2.1	NA	/usr/local/lib/R/site-library
176	harmony	2.0.3	NA	/usr/local/lib/R/site-library
177	haven	2.5.5	NA	/usr/local/lib/R/site-library
178	HDF5Array	1.38.0	NA	/usr/local/lib/R/site-library
179	hdf5r	1.3.12	NA	/usr/local/lib/R/site-library
180	hdWGCNA	0.4.11	NA	/usr/local/lib/R/site-library
181	here	1.0.2	NA	/usr/local/lib/R/site-library
182	hexbin	1.28.5	NA	/usr/local/lib/R/site-library
183	highr	0.12	NA	/usr/local/lib/R/site-library
184	Hmisc	5.2-5	NA	/usr/local/lib/R/site-library
185	hms	1.1.4	NA	/usr/local/lib/R/site-library
186	htmlTable	2.5.0	NA	/usr/local/lib/R/site-library
187	htmltools	0.5.9	NA	/usr/local/lib/R/site-library

	Package	Version	Priority	LibPath
188	htmlwidgets	1.6.4	NA	/usr/local/lib/R/site-library
189	httpuv	1.6.17	NA	/usr/local/lib/R/site-library
190	httr	1.4.8	NA	/usr/local/lib/R/site-library
191	httr2	1.2.2	NA	/usr/local/lib/R/site-library
192	ica	1.0-3	NA	/usr/local/lib/R/site-library
193	ids	1.0.1	NA	/usr/local/lib/R/site-library
194	igraph	2.3.2	NA	/usr/local/lib/R/site-library
195	impute	1.84.0	NA	/usr/local/lib/R/site-library
196	ini	0.3.1	NA	/usr/local/lib/R/site-library
197	IRanges	2.44.0	NA	/usr/local/lib/R/site-library
198	irlba	2.3.7	NA	/usr/local/lib/R/site-library
199	isoband	0.3.0	NA	/usr/local/lib/R/site-library
200	iterators	1.0.14	NA	/usr/local/lib/R/site-library
201	jquerylib	0.1.4	NA	/usr/local/lib/R/site-library
202	jsonlite	2.0.0	NA	/usr/local/lib/R/site-library
203	kableExtra	1.4.0	NA	/usr/local/lib/R/site-library
204	KEGGREST	1.50.0	NA	/usr/local/lib/R/site-library
207	knitr	1.51	NA	/usr/local/lib/R/site-library
208	labeling	0.4.3	NA	/usr/local/lib/R/site-library
209	lambda.r	1.2.4	NA	/usr/local/lib/R/site-library
210	later	1.4.8	NA	/usr/local/lib/R/site-library
212	lazyeval	0.2.3	NA	/usr/local/lib/R/site-library
213	leidenbase	0.1.37	NA	/usr/local/lib/R/site-library
214	lifecycle	1.0.5	NA	/usr/local/lib/R/site-library
215	limma	3.66.0	NA	/usr/local/lib/R/site-library
216	listenv	0.10.1	NA	/usr/local/lib/R/site-library
217	littler	0.3.23	NA	/usr/local/lib/R/site-library
218	lme4	2.0-1	NA	/usr/local/lib/R/site-library
219	lmtest	0.9-40	NA	/usr/local/lib/R/site-library
220	locfit	1.5-9.12	NA	/usr/local/lib/R/site-library
221	lubridate	1.9.5	NA	/usr/local/lib/R/site-library
222	magrittr	2.0.5	NA	/usr/local/lib/R/site-library
223	maps	3.4.3	NA	/usr/local/lib/R/site-library
228	MatrixGenerics	1.22.0	NA	/usr/local/lib/R/site-library
229	MatrixModels	0.5-4	NA	/usr/local/lib/R/site-library
230	matrixStats	1.5.0	NA	/usr/local/lib/R/site-library
231	memoise	2.0.1	NA	/usr/local/lib/R/site-library
234	microbenchmark	1.5.0	NA	/usr/local/lib/R/site-library
235	mime	0.13	NA	/usr/local/lib/R/site-library
236	miniUI	0.1.2	NA	/usr/local/lib/R/site-library
237	minqa	1.2.8	NA	/usr/local/lib/R/site-library
238	modelr	0.1.11	NA	/usr/local/lib/R/site-library
239	monocle3	1.4.27	NA	/usr/local/lib/R/site-library
242	nloptr	2.2.1	NA	/usr/local/lib/R/site-library
244	numDeriv	2016.8-1.1	NA	/usr/local/lib/R/site-library
245	openssl	2.4.1	NA	/usr/local/lib/R/site-library
246	org.At.tair.db	3.22.0	NA	/usr/local/lib/R/site-library
247	otel	0.2.0	NA	/usr/local/lib/R/site-library
248	pak	0.9.5	NA	/usr/local/lib/R/site-library
250	parallelly	1.47.0	NA	/usr/local/lib/R/site-library
251	patchwork	1.3.2	NA	/usr/local/lib/R/site-library
252	pbapply	1.7-4	NA	/usr/local/lib/R/site-library

	Package	Version	Priority	LibPath
253	pbkrtest	0.5.5	NA	/usr/local/lib/R/site-library
254	pbmcapply	1.5.1	NA	/usr/local/lib/R/site-library
255	pheatmap	1.0.13	NA	/usr/local/lib/R/site-library
256	pillar	1.11.1	NA	/usr/local/lib/R/site-library
257	pkgbuild	1.4.8	NA	/usr/local/lib/R/site-library
258	pkgconfig	2.0.3	NA	/usr/local/lib/R/site-library
259	pkgdown	2.2.0	NA	/usr/local/lib/R/site-library
260	pkgload	1.5.2	NA	/usr/local/lib/R/site-library
261	plotly	4.12.0	NA	/usr/local/lib/R/site-library
262	plyr	1.8.9	NA	/usr/local/lib/R/site-library
263	png	0.1-9	NA	/usr/local/lib/R/site-library
264	polyclip	1.10-7	NA	/usr/local/lib/R/site-library
265	polynom	1.4-1	NA	/usr/local/lib/R/site-library
266	praise	1.0.0	NA	/usr/local/lib/R/site-library
267	preprocessCore	1.72.0	NA	/usr/local/lib/R/site-library
268	prettyunits	1.2.0	NA	/usr/local/lib/R/site-library
269	processx	3.9.0	NA	/usr/local/lib/R/site-library
270	profvis	0.4.0	NA	/usr/local/lib/R/site-library
271	progress	1.2.3	NA	/usr/local/lib/R/site-library
272	progressr	0.19.0	NA	/usr/local/lib/R/site-library
273	promises	1.5.0	NA	/usr/local/lib/R/site-library
274	proxy	0.4-29	NA	/usr/local/lib/R/site-library
275	ps	1.9.3	NA	/usr/local/lib/R/site-library
276	pscl	1.5.9	NA	/usr/local/lib/R/site-library
277	purrr	1.2.2	NA	/usr/local/lib/R/site-library
278	quadprog	1.5-8	NA	/usr/local/lib/R/site-library
279	quantreg	6.1	NA	/usr/local/lib/R/site-library
280	qvalue	2.42.0	NA	/usr/local/lib/R/site-library
281	R.methodsS3	1.8.2	NA	/usr/local/lib/R/site-library
282	R.oo	1.27.1	NA	/usr/local/lib/R/site-library
283	R.utils	2.13.0	NA	/usr/local/lib/R/site-library
284	R6	2.6.1	NA	/usr/local/lib/R/site-library
285	ragg	1.5.2	NA	/usr/local/lib/R/site-library
286	RANN	2.6.2	NA	/usr/local/lib/R/site-library
287	rappdirs	0.3.4	NA	/usr/local/lib/R/site-library
288	rbibutils	2.4.1	NA	/usr/local/lib/R/site-library
289	rcmdcheck	1.4.0	NA	/usr/local/lib/R/site-library
290	RColorBrewer	1.1-3	NA	/usr/local/lib/R/site-library
291	Rcpp	1.1.1-1.1	NA	/usr/local/lib/R/site-library
292	RcppAnnoy	0.0.23	NA	/usr/local/lib/R/site-library
293	RcppArmadillo	15.2.7-1	NA	/usr/local/lib/R/site-library
294	RcppEigen	0.3.4.0.2	NA	/usr/local/lib/R/site-library
295	RcppHNSW	0.7.0	NA	/usr/local/lib/R/site-library
296	RcppML	0.3.7.1	NA	/usr/local/lib/R/site-library
297	RcppProgress	0.4.2	NA	/usr/local/lib/R/site-library
298	RcppRoll	0.3.2	NA	/usr/local/lib/R/site-library
299	RcppTOML	0.2.3	NA	/usr/local/lib/R/site-library
300	Rdpack	2.6.6	NA	/usr/local/lib/R/site-library
301	readr	2.2.0	NA	/usr/local/lib/R/site-library
302	readxl	1.5.0	NA	/usr/local/lib/R/site-library
303	reformulas	0.4.4	NA	/usr/local/lib/R/site-library
304	rematch	2.0.0	NA	/usr/local/lib/R/site-library

	Package	Version	Priority	LibPath
305	rematch2	2.1.2	NA	/usr/local/lib/R/site-library
306	remotes	2.5.0	NA	/usr/local/lib/R/site-library
307	reprex	2.1.1	NA	/usr/local/lib/R/site-library
308	reshape2	1.4.5	NA	/usr/local/lib/R/site-library
309	ResidualMatrix	1.20.0	NA	/usr/local/lib/R/site-library
310	reticulate	1.46.0	NA	/usr/local/lib/R/site-library
311	rhdf5	2.54.1	NA	/usr/local/lib/R/site-library
312	rhdf5filters	1.22.0	NA	/usr/local/lib/R/site-library
313	Rhdf5lib	1.32.0	NA	/usr/local/lib/R/site-library
314	RhpcBLASctl	0.23-42	NA	/usr/local/lib/R/site-library
315	Rhtslib	3.6.0	NA	/usr/local/lib/R/site-library
316	rjson	0.2.23	NA	/usr/local/lib/R/site-library
317	rlang	1.2.0	NA	/usr/local/lib/R/site-library
318	rmarkdown	2.31	NA	/usr/local/lib/R/site-library
319	ROCR	1.0-12	NA	/usr/local/lib/R/site-library
320	roxygen2	8.0.0	NA	/usr/local/lib/R/site-library
322	rprojroot	2.1.1	NA	/usr/local/lib/R/site-library
323	rsample	1.3.2	NA	/usr/local/lib/R/site-library
324	Rsamtools	2.26.0	NA	/usr/local/lib/R/site-library
325	RSpectra	0.16-2	NA	/usr/local/lib/R/site-library
326	RSQLite	3.53.1	NA	/usr/local/lib/R/site-library
327	rstatix	0.7.3	NA	/usr/local/lib/R/site-library
328	rstudioapi	0.18.0	NA	/usr/local/lib/R/site-library
329	rsvd	1.0.5	NA	/usr/local/lib/R/site-library
330	Rtsne	0.17	NA	/usr/local/lib/R/site-library
331	rversions	3.0.0	NA	/usr/local/lib/R/site-library
332	rvest	1.0.5	NA	/usr/local/lib/R/site-library
333	s2	1.1.11	NA	/usr/local/lib/R/site-library
334	S4Arrays	1.10.1	NA	/usr/local/lib/R/site-library
335	S4Vectors	0.48.1	NA	/usr/local/lib/R/site-library
336	S7	0.2.2	NA	/usr/local/lib/R/site-library
337	sass	0.4.10	NA	/usr/local/lib/R/site-library
338	ScaledMatrix	1.18.0	NA	/usr/local/lib/R/site-library
339	scales	1.4.0	NA	/usr/local/lib/R/site-library
340	scater	1.38.1	NA	/usr/local/lib/R/site-library
341	scattermore	1.2	NA	/usr/local/lib/R/site-library
342	scatterpie	0.2.6	NA	/usr/local/lib/R/site-library
343	sctransform	0.4.3	NA	/usr/local/lib/R/site-library
344	scuttle	1.20.0	NA	/usr/local/lib/R/site-library
345	selectr	0.5-1	NA	/usr/local/lib/R/site-library
346	Seqinfo	1.0.0	NA	/usr/local/lib/R/site-library
347	sessioninfo	1.2.3	NA	/usr/local/lib/R/site-library
348	Seurat	5.5.0	NA	/usr/local/lib/R/site-library
349	SeuratDisk	0.0.0.9021	NA	/usr/local/lib/R/site-library
350	SeuratObject	5.4.0	NA	/usr/local/lib/R/site-library
351	SeuratWrappers	0.4.0	NA	/usr/local/lib/R/site-library
352	sf	1.1-1	NA	/usr/local/lib/R/site-library
353	shape	1.4.6.1	NA	/usr/local/lib/R/site-library
354	shiny	1.13.0	NA	/usr/local/lib/R/site-library
355	SingleCellExperiment	1.32.0	NA	/usr/local/lib/R/site-library
356	sitmo	2.0.2	NA	/usr/local/lib/R/site-library
357	slam	0.1-55	NA	/usr/local/lib/R/site-library

	Package	Version	Priority	LibPath
358	slider	0.3.3	NA	/usr/local/lib/R/site-library
359	snow	0.4-4	NA	/usr/local/lib/R/site-library
360	sourcetools	0.1.7-2	NA	/usr/local/lib/R/site-library
361	sp	2.2-1	NA	/usr/local/lib/R/site-library
362	spam	2.11-4	NA	/usr/local/lib/R/site-library
363	SparseArray	1.10.10	NA	/usr/local/lib/R/site-library
364	SparseM	1.84-2	NA	/usr/local/lib/R/site-library
365	sparseMatrixStats	1.22.0	NA	/usr/local/lib/R/site-library
367	spatstat.data	3.1-9	NA	/usr/local/lib/R/site-library
368	spatstat.explore	3.8-1	NA	/usr/local/lib/R/site-library
369	spatstat.geom	3.8-1	NA	/usr/local/lib/R/site-library
370	spatstat.random	3.5-0	NA	/usr/local/lib/R/site-library
371	spatstat.sparse	3.2-0	NA	/usr/local/lib/R/site-library
372	spatstat.univar	3.2-0	NA	/usr/local/lib/R/site-library
373	spatstat.utils	3.2-3	NA	/usr/local/lib/R/site-library
374	spData	2.3.5	NA	/usr/local/lib/R/site-library
375	spdep	1.4-2	NA	/usr/local/lib/R/site-library
376	speedglm	0.3-4	NA	/usr/local/lib/R/site-library
378	statmod	1.5.2	NA	/usr/local/lib/R/site-library
381	stringi	1.8.7	NA	/usr/local/lib/R/site-library
382	stringr	1.6.0	NA	/usr/local/lib/R/site-library
383	SummarizedExperiment	1.40.0	NA	/usr/local/lib/R/site-library
386	svglite	2.2.2	NA	/usr/local/lib/R/site-library
387	sys	3.4.3	NA	/usr/local/lib/R/site-library
388	systemfonts	1.3.2	NA	/usr/local/lib/R/site-library
390	tensor	1.5.1	NA	/usr/local/lib/R/site-library
391	tester	0.3.0	NA	/usr/local/lib/R/site-library
392	testthat	3.3.2	NA	/usr/local/lib/R/site-library
393	textshaping	1.0.5	NA	/usr/local/lib/R/site-library
394	tibble	3.3.1	NA	/usr/local/lib/R/site-library
395	tidydr	0.0.6	NA	/usr/local/lib/R/site-library
396	tidygraph	1.3.1	NA	/usr/local/lib/R/site-library
397	tidyr	1.3.2	NA	/usr/local/lib/R/site-library
398	tidyselect	1.2.1	NA	/usr/local/lib/R/site-library
399	tidytree	0.4.7	NA	/usr/local/lib/R/site-library
400	tidyverse	2.0.0	NA	/usr/local/lib/R/site-library
401	timechange	0.4.0	NA	/usr/local/lib/R/site-library
402	timeDate	4052.112	NA	/usr/local/lib/R/site-library
403	tinytex	0.59	NA	/usr/local/lib/R/site-library
405	treeio	1.34.0	NA	/usr/local/lib/R/site-library
406	tweenr	2.0.3	NA	/usr/local/lib/R/site-library
407	tzdb	0.5.0	NA	/usr/local/lib/R/site-library
408	UCell	2.14.0	NA	/usr/local/lib/R/site-library
409	UCSC.utils	1.6.1	NA	/usr/local/lib/R/site-library
410	units	1.0-1	NA	/usr/local/lib/R/site-library
411	UpSetR	1.4.1	NA	/usr/local/lib/R/site-library
412	urca	1.3-4	NA	/usr/local/lib/R/site-library
413	urlchecker	1.0.1	NA	/usr/local/lib/R/site-library
414	usethis	3.2.1	NA	/usr/local/lib/R/site-library
415	utf8	1.2.6	NA	/usr/local/lib/R/site-library
417	uuid	1.2-2	NA	/usr/local/lib/R/site-library
418	uwot	0.2.4	NA	/usr/local/lib/R/site-library

	Package	Version	Priority	LibPath
419	vctrs	0.7.3	NA	/usr/local/lib/R/site-library
420	VennDiagram	1.8.2	NA	/usr/local/lib/R/site-library
421	vipor	0.4.7	NA	/usr/local/lib/R/site-library
422	viridis	0.6.5	NA	/usr/local/lib/R/site-library
423	viridisLite	0.4.3	NA	/usr/local/lib/R/site-library
424	vroom	1.7.1	NA	/usr/local/lib/R/site-library
425	waldo	0.6.2	NA	/usr/local/lib/R/site-library
426	warp	0.2.3	NA	/usr/local/lib/R/site-library
427	WGCNA	1.74	NA	/usr/local/lib/R/site-library
428	whisker	0.4.1	NA	/usr/local/lib/R/site-library
429	withr	3.0.2	NA	/usr/local/lib/R/site-library
430	wk	0.9.5	NA	/usr/local/lib/R/site-library
431	WriteXLS	6.8.0	NA	/usr/local/lib/R/site-library
432	xfun	0.58	NA	/usr/local/lib/R/site-library
433	xml2	1.5.2	NA	/usr/local/lib/R/site-library
434	xopen	1.0.1	NA	/usr/local/lib/R/site-library
435	xtable	1.8-8	NA	/usr/local/lib/R/site-library
436	XVector	0.50.0	NA	/usr/local/lib/R/site-library
437	yaml	2.3.12	NA	/usr/local/lib/R/site-library
438	yulab.utils	0.2.4	NA	/usr/local/lib/R/site-library
439	zellkonverter	1.20.1	NA	/usr/local/lib/R/site-library
440	zip	2.3.3	NA	/usr/local/lib/R/site-library
441	zoo	1.8-15	NA	/usr/local/lib/R/site-library

Table 4: Complete Python package inventory in the Docker image.

Package	Version
adjustText	1.3.0
anndata	0.12.16
annotated-types	0.7.0
anyio	4.13.0
argon2-cffi	25.1.0
argon2-cffi-bindings	25.1.0
array-api-compat	1.14.0
arrow	1.4.0
asttokens	3.0.1
async-lru	2.3.0
attrs	26.1.0
babel	2.18.0
beautifulsoup4	4.14.3
bleach	6.3.0
blosc2	4.4.1
cellbender	0.3.2
certifi	2026.5.20
cffi	2.0.0
charset-normalizer	3.4.7
click	8.4.1
comm	0.2.3
contourpy	1.3.3
cuda-bindings	13.3.1

Package	Version
cuda-pathfinder	1.5.5
cuda-toolkit	13.0.2
cycler	0.12.1
debugpy	1.8.21
decorator	5.3.1
defusedxml	0.7.1
donfig	0.8.1.post1
elpigraph-python	0.3.4
entrypoints	0.4
executing	2.2.1
fa2_modified	0.4
fast-array-utils	1.4.1
fastjsonschema	2.21.2
filelock	3.29.0
fonttools	4.63.0
fqdn	1.5.1
fsspec	2026.4.0
google-crc32c	1.8.0
h11	0.16.0
h5py	3.16.0
httpcore	1.0.9
httpx	0.28.1
idna	3.18
igraph	1.0.0
ipykernel	6.29.5
ipython	9.14.0
ipython-genutils	0.2.0
ipython_pygments_lexers	1.1.1
ipywidgets	8.1.8
isoduration	20.11.0
jax	0.10.1
jaxlib	0.10.1
jaxopt	0.8.5
jedi	0.20.0
Jinja2	3.1.6
joblib	1.5.3
json5	0.14.0
jsonpointer	3.1.1
jsonschema	4.26.0
jsonschema-specifications	2025.9.1
jupyter	1.1.1
jupyter-console	6.6.3
jupyter-events	0.12.1
jupyter-highlight-selected-word	0.2.0
jupyter-lsp	2.3.1
jupyter_client	7.4.9
jupyter_contrib_core	0.4.2
jupyter_contrib_nbextensions	0.7.0
jupyter_core	5.9.1
jupyter_nbextensions_configurator	0.6.4
jupyter_server	2.19.0
jupyter_server_terminals	0.5.4

Package	Version
jupyterlab	4.5.7
jupyterlab_pygments	0.3.0
jupyterlab_server	2.28.0
jupyterlab_widgets	3.0.16
kiwisolver	1.5.0
lark	1.3.1
legacy-api-wrap	1.5
leidenalg	0.12.0
llvmlite	0.47.0
loompy	3.0.8
lxml	6.1.1
lxml_html_clean	0.4.5
MarkupSafe	3.0.3
matplotlib	3.10.9
matplotlib-inline	0.2.2
mellon	1.7.1
mistune	0.8.4
ml_dtypes	0.5.4
mpmath	1.3.0
msgpack	1.1.2
narwhals	2.22.0
natsort	8.4.0
nbclassic	1.3.3
nbclient	0.10.4
nbconvert	6.5.4
nbformat	5.10.4
ndindex	1.10.1
nest-asyncio	1.6.0
networkx	3.6.1
notebook	6.5.7
notebook_shim	0.2.4
numba	0.65.1
numcodecs	0.16.5
numexpr	2.14.1
numpy	2.4.6
numpy-groupies	0.11.3
nvidia-cublas	13.1.1.3
nvidia-cuda-cupti	13.0.85
nvidia-cuda-nvrtc	13.0.88
nvidia-cuda-runtime	13.0.96
nvidia-cudnn-cu13	9.20.0.48
nvidia-cufft	12.0.0.61
nvidia-cufile	1.15.1.6
nvidia-curand	10.4.0.35
nvidia-cusolver	12.0.4.66
nvidia-cusparse	12.6.3.3
nvidia-cusparselt-cu13	0.8.1
nvidia-nccl-cu13	2.29.7
nvidia-nvjitlink	13.0.88
nvidia-nvshmem-cu13	3.4.5
nvidia-nvtx	13.0.85
opt_einsum	3.4.0

Package	Version
packaging	26.2
palantir	1.4.4
pandas	2.3.3
pandocfilters	1.5.1
parso	0.8.7
patsy	1.0.2
pexpect	4.9.0
pillow	12.2.0
pip	26.1.2
platformdirs	4.10.0
plotly	6.7.0
prometheus_client	0.25.0
prompt_toolkit	3.0.52
psutil	7.2.2
ptyprocess	0.7.0
pure_eval	0.2.3
py-cpuinfo	9.0.0
pycparser	3.0
pydantic	2.13.4
pydantic_core	2.46.4
Pygments	2.20.0
pynndescent	0.6.0
pyparsing	3.3.2
pyro-api	0.1.2
pyro-ppl	1.9.1
python-dateutil	2.9.0.post0
python-igraph	1.0.0
python-json-logger	4.1.0
pytz	2026.2
PyYAML	6.0.3
pyzmq	27.1.0
referencing	0.37.0
requests	2.34.2
rfc3339-validator	0.1.4
rfc3986-validator	0.1.1
rfc3987-syntax	1.1.0
rpds-py	2026.5.1
rpy2	3.6.7
rpy2-rinterface	3.6.6
rpy2-robjects	3.6.5
scanpy	1.12.1
scFates	1.2.4
scikit-learn	1.9.0
scikit-misc	0.5.2
scipy	1.17.1
scverse-misc	0.0.7
seaborn	0.13.2
Send2Trash	2.1.0
session-info2	0.4.1
setuptools	81.0.0
shapely	2.1.2
simpleppt	1.1.7

Package	Version
six	1.17.0
soupsieve	2.8.4
stack-data	0.6.3
statsmodels	0.14.6
sympy	1.14.0
tables	3.11.1
terminado	0.18.1
texttable	1.7.0
threadpoolctl	3.6.0
tinycss2	1.5.1
torch	2.12.0
tornado	6.5.6
tqdm	4.67.3
traitlets	5.15.0
triton	3.7.0
typing-inspection	0.4.2
typing_extensions	4.15.0
tzdata	2026.2
tzlocal	5.4.3
umap-learn	0.5.12
uri-template	1.3.0
urllib3	2.7.0
wcwidth	0.7.0
webcolors	25.10.0
webencodings	0.5.1
websocket-client	1.9.0
widgetsnbextension	4.0.15
zarr	3.2.1